- The acceleration due to gravity is denoted by $g\,\text{m}\,\text{s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

4 A ruler PQRS is a uniform rectangular lamina with mass 20 grams. The length of PQ is 30 cm and the length of PS is 4 cm. The ruler is attached at P to a smooth hinge and held with S vertically below P by a horizontal force of magnitude F N as shown in the diagram.



(a) Calculate the value of F. **[3]**

(b) Explain what would happen to the lamina if the force at S were removed. **[1]**